

L11 — Servlet CRUD — District Quiz Game

11.1 Theory — Database Schema

Question Table

Field	Type	Description
question_id	INT	Primary Key
question_text	VARCHAR	Question
category	VARCHAR	Crop/Geography/Institution
option_a	VARCHAR	Option A
option_b	VARCHAR	Option B
option_c	VARCHAR	Option C
option_d	VARCHAR	Option D
correct_answer	CHAR	Correct Answer

PlayerScore Table

Field	Type	Description
player_id	INT	Primary Key
player_name	VARCHAR	Player Name
final_score	INT	Final Score

Justification: Questions and player scores are stored separately to keep the database organized. The category field identifies the type of question.

11.2 Practical — Save Player Score

```
import java.sql.*;

public class PlayerScoreDAO {

    public void saveScore(String name, int score) {

        String sql =

            "INSERT INTO PlayerScore(player_name, final_score) VALUES (?, ?)";
```

```

try (Connection con = DatabaseConnection.getConnection());

    PreparedStatement ps = con.prepareStatement(sql)) {

    ps.setString(1, name);

    ps.setInt(2, score);

    ps.executeUpdate();

} catch (Exception e) {

    e.printStackTrace();

}

}

}

```

Database Connection

```

public class DatabaseConnection {

    public static Connection getConnection() throws Exception {

        return DriverManager.getConnection(

            "jdbc:mysql://localhost:3306/quizdb",

            "root",

            "your_password"

        );

    }

}

```

11.3 Practical — Quiz Logic

Question Class

```

class Question {

    String question;

    String[] options;

    int correctAnswer;

```

```

Question(String q, String[] o, int a) {
    question = q;
    options = o;
    correctAnswer = a;
}

boolean checkAnswer(int answer) {
    return answer == correctAnswer;
}
}

```

Quiz Logic

```

Question[] questions = {
    new Question(
        "Which is a common crop in Bangladesh?",
        new String[]{"Tea", "Coffee", "Olive", "Apple"}, 1
    ),
    new Question(
        "Which district has Madhupur Garh?",
        new String[]{"Tangail", "Khulna", "Sylhet", "Rajshahi"}, 1
    ),
    new Question(
        "Which university is located in Tangail?",
        new String[]{"MBSTU", "BUET", "CUET", "RUET"}, 1
    )
};

int score = 0;

for (Question q : questions) {

```

```
int answer = 1;

if (q.checkAnswer(answer)) {

    score++;

}

}

System.out.println("Final Score: " + score);
```

Result: The quiz presents MCQs, checks answers, and increases the score for every correct answer.